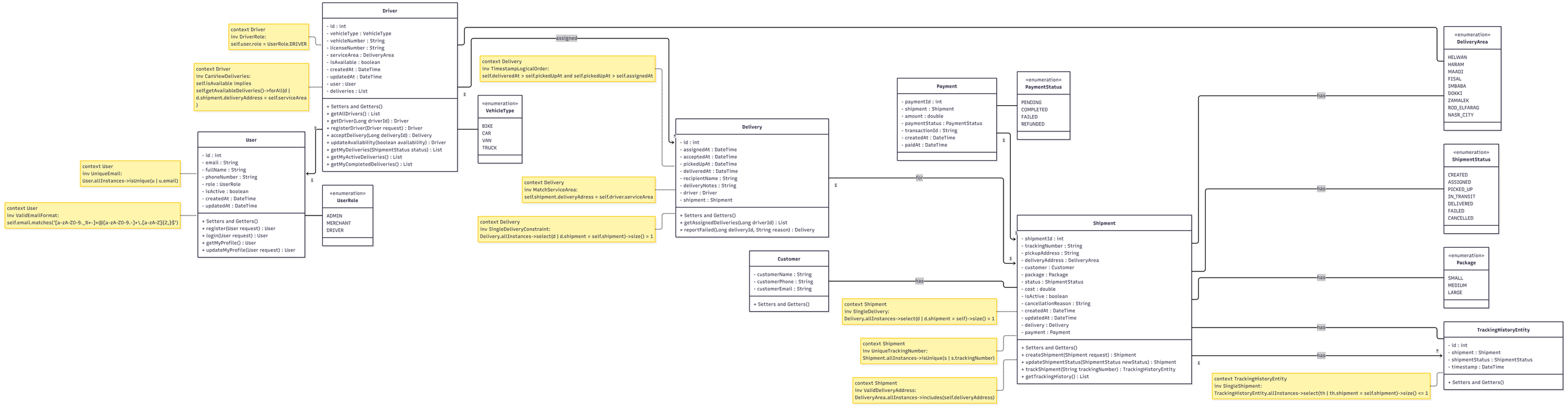
Driver
- id : int
- vehicleType : VehicleType
- vehicleNumber : String
- licenseNumber : String
- serviceArea : DeliveryArea
- isAvailable : boolean
- createdAt : DateTime
- updatedAt : DateTime
- user : User
- deliveries : List
+ Setters and Getters()
+ getAllDrivers() : List
+ getDriver(Long driverId) : Driver
+ registerDriver(Driver request) : Driver
+ acceptDelivery(Long deliveryId) : Delivery
+ updateAvailability(boolean availability) : Driver
+ getMyDeliveries(ShipmentStatus status) : List
+ getMyActiveDeliveries() : List
+ getMyCompletedDeliveries() : List
context Driver
inv DriverRole:
self.user.role = UserRole.DRIVER
context Driver
inv CanViewDeliveries:
self.isAvailable implies
self.getAvailableDeliveries()->forAll(d |
d.shipment.deliveryAddress = self.serviceArea
)
assigned
1
context Delivery
inv TimestampLogicalOrder:
self.deliveredAt > self.pickedUpAt and self.pickedUpAt > self.assignedAt
«enumeration»
VehicleType
BIKE
CAR
VAN
TRUCK
Delivery
- id : int
- assignedAt : DateTime
- acceptedAt : DateTime
- pickedUpAt : DateTime
- deliveredAt : DateTime
- recipientName : String
- deliveryNotes : String
- driver : Driver
- shipment : Shipment
+ Setters and Getters()
+ getAssignedDeliveries(Long driverId) : List
+ reportFailed(Long deliveryId, String reason) : Delivery
1
for
1
User
- id : int
- email : String
- fullName : String
- phoneNumber : String
- role : UserRole
- isActive : boolean
- createdAt : DateTime
- updatedAt : DateTime
+ Setters and Getters()
+ register(User request) : User
+ login(User request) : User
+ getMyProfile() : User
+ updateMyProfile(User request) : User
context User
inv UniqueEmail:
User.allInstances->isUnique(u | u.email)
context User
inv ValidEmailFormat:
self.email.matches('[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$')
«enumeration»
UserRole
ADMIN
MERCHANT
DRIVER
context Delivery
inv MatchServiceArea:
self.shipment.deliveryAdress = self.driver.serviceArea
context Delivery
inv SingleDeliveryConstraint:
Delivery.allInstances->select(d | d.shipment = self.shipment)->size() = 1
Payment
- paymentId : int
- shipment : Shipment
- amount : double
- paymentStatus : PaymentStatus
- transactionId : String
- createdAt : DateTime
- paidAt : DateTime
«enumeration»
PaymentStatus
PENDING
COMPLETED
FAILED
REFUNDED
Customer
- customerName : String
- customerPhone : String
- customerEmail : String
+ Setters and Getters()
has
Shipment
- shipmentId : int
- trackingNumber : String
- pickupAddress : String
- deliveryAddress : DeliveryArea
- customer : Customer
- package : Package
- status : ShipmentStatus
- cost : double
- isActive : boolean
- cancellationReason : String
- createdAt : DateTime
- updatedAt : DateTime
- delivery : Delivery
- payment : Payment
+ Setters and Getters()
+ createShipment(Shipment request) : Shipment
+ updateShipmentStatus(ShipmentStatus newStatus) : Shipment
+ trackShipment(String trackingNumber) : TrackingHistoryEntity
+ getTrackingHistory() : List
context Shipment
inv SingleDelivery:
Delivery.allInstances->select(d | d.shipment = self)->size() = 1
context Shipment
inv UniqueTrackingNumber:
Shipment.allInstances->isUnique(s | s.trackingNumber)
context Shipment
inv ValidDeliveryAddress:
DeliveryArea.allInstances->includes(self.deliveryAddress)
has
«enumeration»
DeliveryArea
HELWAN
HARAM
MAADI
FISAL
IMBABA
DOKKI
ZAMALEK
ROD_ELFARAG
NASR_CITY
«enumeration»
ShipmentStatus
CREATED
ASSIGNED
PICKED_UP
IN_TRANSIT
DELIVERED
FAILED
CANCELLED
«enumeration»
Package
SMALL
MEDIUM
LARGE
TrackingHistoryEntity
- id : int
- shipment : Shipment
- shipmentStatus : ShipmentStatus
- timestamp : DateTime
+ Setters and Getters()
context TrackingHistoryEntity
inv SingleShipment:
TrackingHistoryEntity.allInstances->select(th | th.shipment = self.shipment)->size() <= 1